COMPUTER NETWORKS

CSE 321

IoT Networking Challenges

- Allowing communication among various devices simultaneously.

IoT Networking Challenges

- Allowing communication among various devices simultaneously.
- Ensuring security in critical areas like health, industry, home surveillance etc.

IoT Networking Challenges

- Allowing communication among various devices simultaneously.
- Ensuring security in critical areas like health, industry, home surveillance etc.
- Being efficient (time, memory, energy etc).

IoT Networking Challenges

- Allowing communication among various devices simultaneously.
- Ensuring security in critical areas like health, industry, home surveillance etc.
- Being efficient (time, memory, energy etc).
- Providing scalability.

Choosing A Protocol

Depends on requirements, environment and task

Choosing A Protocol

Depends on requirements, environment and task

Need to consider:

- Ranges (small / wide)
- Data Rates (low / high)
- Power Consumption
- Cost

Common Wireless IoT Protocols

Protocols	Range	Data Rate	Energy
Bluetooth & BLE	Short	Low	Low

Common Wireless IoT Protocols

Protocols	Range	Data Rate	Energy
Bluetooth & BLE	Short	Low	Low
Zigbee	Short	Low	Low

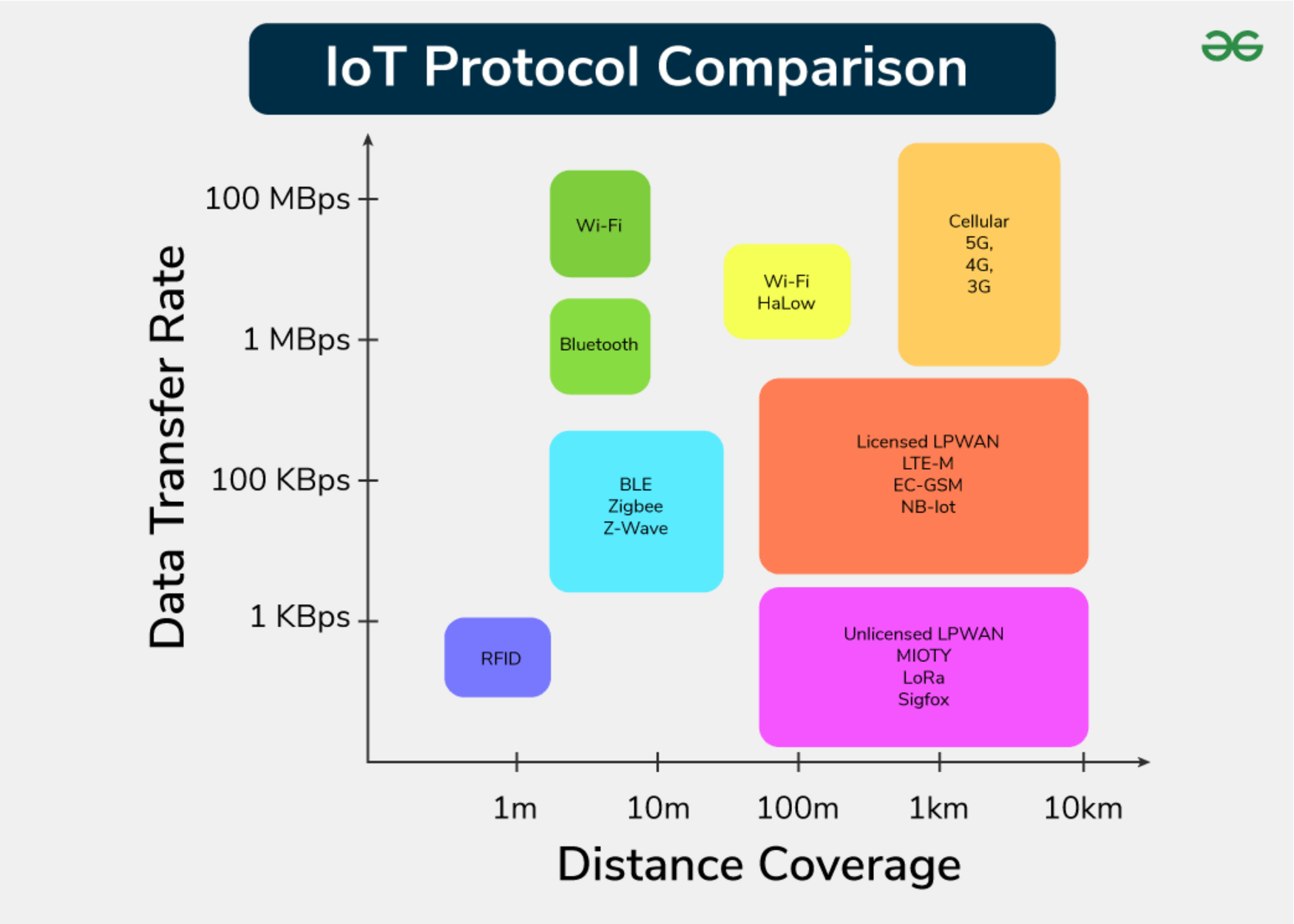
Common Wireless IoT Protocols

Protocols	Range	Data Rate	Energy
Bluetooth & BLE	Short	Low	Low
Zigbee	Short	Low	Low
Wi-Fi	Short	High	High

Common Wireless IoT Protocols

Protocols	Range	Data Rate	Energy
Bluetooth & BLE	Short	Low	Low
Zigbee	Short	Low	Low
Wi-Fi	Short	High	High
LoRaWAN	Long	Low	Low

Common Wireless IoT Protocols



IoT Network Layer Protocols

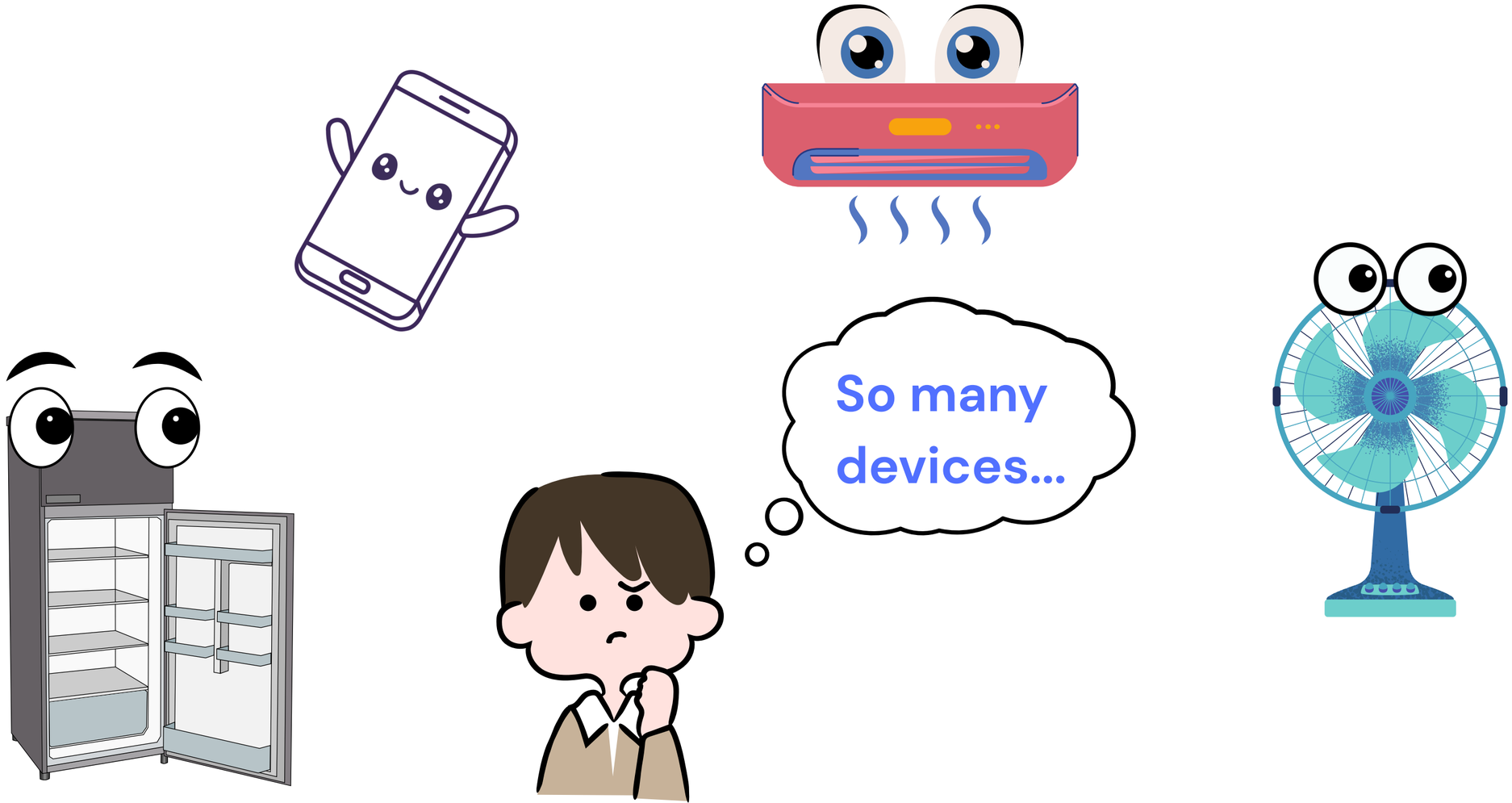
- RPL
- IPv4
- IPv6

IPv4 to IPv6

Version 4 → 2^32 only

IPv4 to IPv6

Version 4 → 2^32 only



IPv4 to IPv6

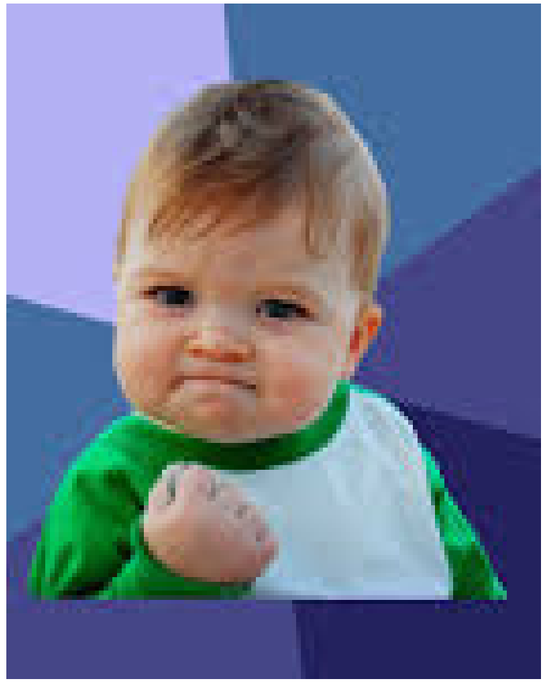
Version 4 → 2^32

Version 6 → 2^128

IoT is one of the major drives

Common Application Layer Protocols

- MQTT
- HTTP/HTTPS
- CoAP
- DDS



End of Slide